

Smog City: Visualizing the Invisible Threat

Air Quality Analysis of Indian Cities

1. Introduction

Air pollution is one of the major environmental and public-health challenges in India. Poor air quality can affect respiratory and cardiovascular health, reduce visibility, and decrease the overall quality of life. This project focuses on analysing Air Quality Index (AQI) data from major Indian cities to understand pollution trends, seasonal variations, and the pollutants responsible for changes in air quality.

2. Project Title

Smog City: Visualizing the Invisible Threat – Air Quality Analysis of Indian Cities

The project belongs to the domain of **Environment, Climate & Energy** and is developed as part of the Data Science and Visualization course (24ADI204).

3. Problem Statement

Air pollution levels vary across different Indian cities and time periods. Analysing this data can help identify highly polluted cities, seasonal pollution patterns, and the major pollutants affecting AQI. The project aims to transform raw AQI observations into meaningful information that can support public awareness and informed decision-making.

4. Objectives

The main objectives of the project are to perform city-level and time-based AQI analysis and identify pollution trends and extreme pollution events. The project also aims to handle missing values and outliers, identify important pollutants such as PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃ and VOCs, and use Principal Component Analysis (PCA) to understand relationships between pollutants. An interactive dashboard will also be developed to communicate the findings clearly.

5. Dataset Description

The project uses the “**Air Quality Data in India (2015–2020)**” dataset obtained from Kaggle. The primary file used is **city_day.csv**. The dataset contains multi-year daily observations from major Indian cities. Important features include City, Date, AQI, PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃, Benzene, Toluene and AQI Category.

6. Data Preprocessing

The dataset contains missing values and inconsistencies, making data preprocessing an important part of the project. The preprocessing stage includes validating the collected data, parsing dates, normalising city names, handling missing values using suitable imputation strategies, and detecting outliers using statistical and domain-based methods.

7. Exploratory Data Analysis

Exploratory Data Analysis (EDA) will be performed to identify trends, seasonal patterns, pollution extremes and differences between cities. The analysis will include AQI trends over multiple years, city-wise pollution comparisons, pollutant relationships and correlation analysis. These analyses help identify pollution hotspots and understand how different pollutants vary together.

8. Principal Component Analysis

Principal Component Analysis (PCA) will be used to reduce the dimensionality of the pollutant variables. Since several pollutants may be correlated, PCA can combine their information into principal components. The project will examine the component loadings to understand possible pollution sources and use PCA scores to identify similarities between cities based on their pollutant patterns.

9. Visualization and Dashboard

An interactive dashboard will be developed using Plotly and Streamlit. The dashboard will present AQI trends, city-wise comparisons, correlation heatmaps and an interactive map of India. A time slider and city filters will allow users to explore pollution patterns across different locations and periods.

10. Tools and Technologies

The project will use **Python, Pandas, NumPy and Scikit-learn** for data analysis and PCA. **Matplotlib and Seaborn** will be used for static visualizations, while **Plotly and Streamlit** will be used to create interactive visualizations and the dashboard. GitHub will be used for version control and reproducibility.

11. Expected Outcomes

The project is expected to identify the most polluted cities and high-risk seasonal periods. It will also determine which pollutants contribute most to AQI variation and reveal pollutant source patterns using correlation analysis and PCA. The final dashboard will allow stakeholders to explore pollution trends and support data-driven recommendations for local mitigation strategies.

12. Conclusion

The **Smog City** project converts raw AQI data into useful and understandable insights through data cleaning, exploratory analysis, PCA and interactive visualization. The project can improve understanding of air pollution patterns across Indian cities and support environmental awareness and informed decision-making. Future work includes refining the imputation process, validating PCA components with source data, and deploying the Streamlit dashboard with city filters and a time slider.